

LAB EXPERIMENT RECORD

Set 13: Geographic Data

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	N LAHARI
Register Number	192473018
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Aim

To visualize geographic data using a map chart, scatter plot, and table, and to create an interactive Tableau dashboard for exploring the relationship between population, average temperature, and elevation. **Dataset**

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

1. Create a Map Chart to Visualize the Distribution of Cities

Aim

To create a map chart to visualize the geographic distribution of cities and label the map elements.

R Code

R

```

city <- c("City A", "City B", "City C")
population <- c(500000, 700000, 600000)
temperature <- c(75, 68, 80)
elevation <- c(1000, 800, 1200)

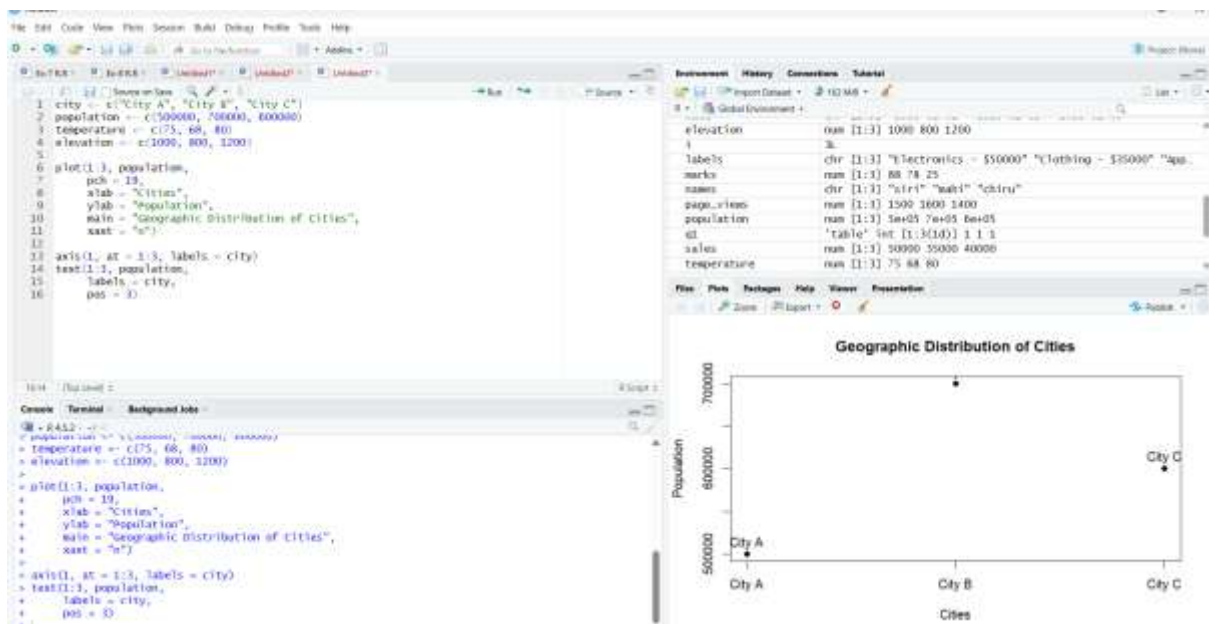
plot(1:3, population,
     pch = 19,
     xlab = "Cities",
     ylab = "Population",
     main = "Geographic Distribution of Cities",
     xaxt = "n")

axis(1, at = 1:3, labels = city)

text(1:3, population,
     labels = city,
     pos = 3)

```

Output



Observation

City B has the highest population, followed by City C and City A.

2. Scatter Plot – Temperature vs Population

Aim

To create a scatter plot to study the relationship between average temperature and population size.

R Code

R

```
city <- c("City A", "City B", "City C")
```

```
population <- c(500000, 700000, 600000)
```

```
temperature <- c(75, 68, 80)
```

```
plot(temperature, population,
```

```
  pch = 19,
```

```
  xlab = "Average Temperature",
```

```
  ylab = "Population",
```

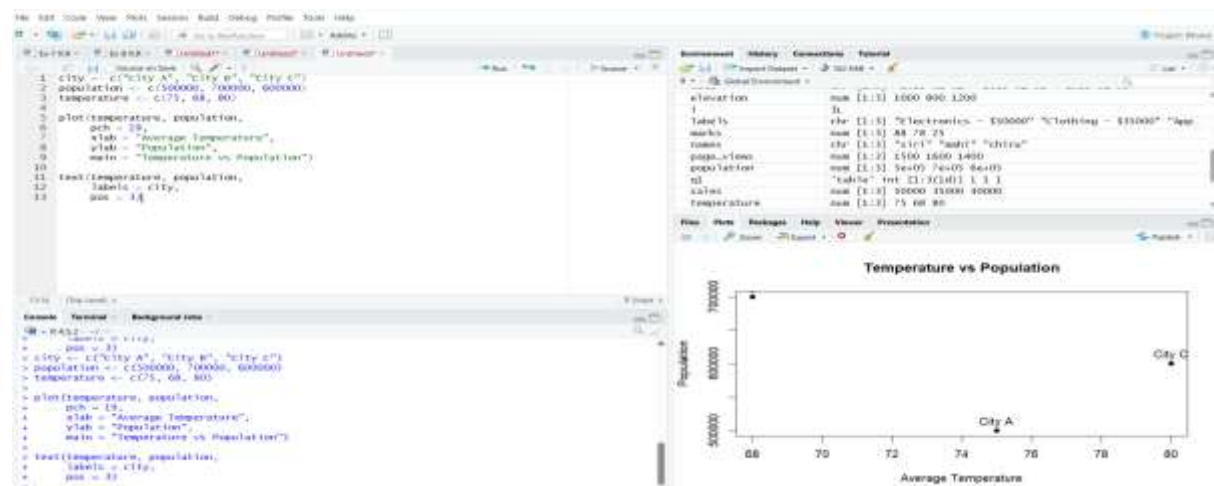
```
  main = "Temperature vs Population")
```

```
text(temperature, population,
```

```
  labels = city,
```

```
  pos = 3)
```

Output



A scatter plot showing the relationship between **Average Temperature** and **Population**.

3. Build a Table to Display Geographic Data

Aim

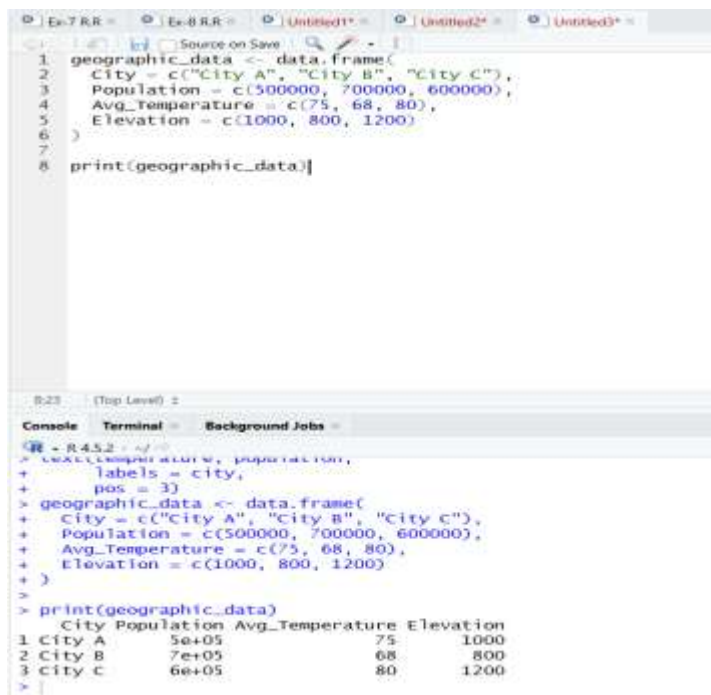
To create a table showing the geographic information of each city.

R Code

R

```
geographic_data <- data.frame(  
  City = c("City A", "City B", "City C"),  
  Population = c(500000, 700000, 600000),  
  Avg_Temperature = c(75, 68, 80),  
  Elevation = c(1000, 800, 1200)  
)  
  
print(geographic_data)
```

Output



```
R 4.5.2 - x64_64  
> geographic_data <- data.frame(  
+   City = c("City A", "City B", "City C"),  
+   Population = c(500000, 700000, 600000),  
+   Avg_Temperature = c(75, 68, 80),  
+   Elevation = c(1000, 800, 1200)  
+ )  
> print(geographic_data)
```

	City	Population	Avg_Temperature	Elevation
1	City A	5e+05	75	1000
2	City B	7e+05	68	800
3	City C	6e+05	80	1200

Observation

The table clearly displays the population, average temperature and elevation of all three cities.

4. Tableau Dashboard

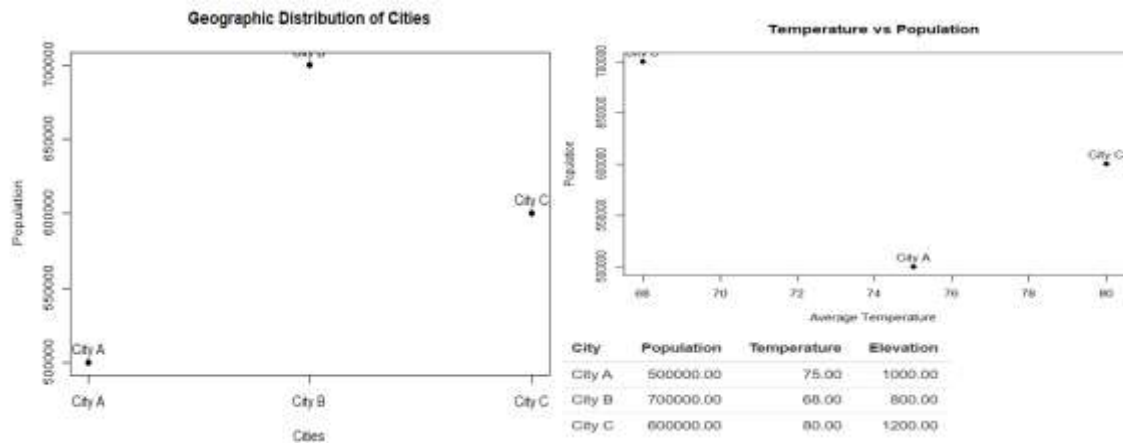
Aim

To develop an interactive dashboard combining the map chart, scatter plot and geographic data table.

Tableau Procedure

1. Open **Tableau**.
2. Import the Geographic Data dataset.
3. Create a **Map Chart** using the City field.
4. Add Population to the map to represent city size.
5. Create a **Scatter Plot**.
6. Drag **Avg. Temperature** to Columns.
7. Drag **Population** to Rows.
8. Add **City** as the label.
9. Create a **Table** using City, Population, Avg. Temperature and Elevation.
10. Select **Dashboard** → **New Dashboard**.
11. Drag the Map Chart, Scatter Plot and Table into the dashboard.
12. Add suitable titles and labels.
13. Save the dashboard.

Output:



Result

The geographic data was successfully analyzed using a map chart, scatter plot and table. The dashboard provides an interactive view of city population, temperature and elevation.